BPT-109 Assignment-1 Master Compilation

Comprehensive Solutions for Questions 1 through 8

Student Name
Muskan Bhasin
Roll Number
1826055

Q1 Generations & Specs | Q2 CPU Architecture | Q3 Campus Automation
Q4 Language Taxonomy | Q5 Spatial Digitizers | Q6 Plotter vs UHD+Printer
Q7 Biometrics Security | Q8 Accessible Workstation

Question 1 (Remembering) Computer Characteristics & Generations

Primary Characteristics of Modern Computers & Hardware Generations

Question Statement: List four primary characteristics of a modern digital computer and identify the key hardware technology associated with each of the five generations of computers.

1. High Speed

Executes billions of floating-point instructions per second (GFLOPS/TFLOPS) with nanosecond/picosecond cycle times.

2. Accuracy & GIGO

Computes with near-absolute mathematical precision. Errors arise from flawed data/programming (Garbage In, Garbage Out).

3. Diligence

Operates continuously without fatigue, boredom, or loss of concentration, maintaining identical fidelity across millions of loops.

4. Versatility & Memory

Executes diverse tasks (scientific, media, AI) using non-volatile storage and RAM hierarchy for instant instruction switching.

The Five Hardware Generations Matrix

1st Gen (1940-1956)

Vacuum Tubes

Thermionic Valves, Magnetic Drums, Machine Code (ENIAC, UNIVAC).

2nd Gen (1956-1963)

Transistors

Semiconductors, Magnetic Core Memory, Assembly/FORTRAN (IBM 7094).

3rd Gen (1964-1971)

Integrated Circuits

SSI/MSI Chips, Keyboards, Monitors, OS Multiprogramming (IBM 360).

4th Gen (1971-Present)

VLSI / Microprocessor

Single-chip Microprocessors, PCs, Networking, GUIs (Intel 4004 to i9).

5th Gen (Present-Beyond)

ULSI / AI & Quantum

Parallel Processing, Neural Chips, NLP, Quantum Qubits.

Interactive Hardware Architecture Selector

Gen 1: Vacuum Tube | Gen 2: Transistor | Gen 3: IC | Gen 4: VLSI Microprocessor | Gen 5: ULSI & Quantum

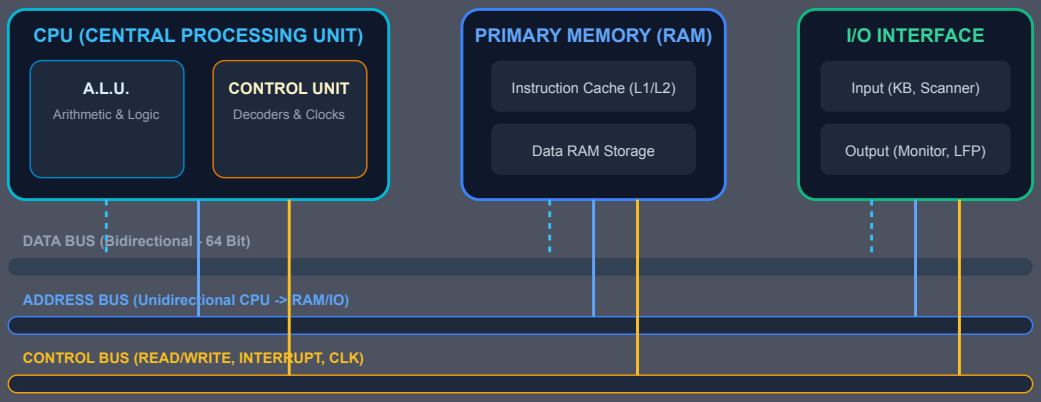
Question 2 (Understanding) System Architecture & Signal Flow

Computer System Architecture & Signal Bus Flow

Question Statement: Explain the flow of data, instructions, and control signals within the functional block diagram of a computer, detailing the interaction between

the CPU, memory unit, and I/O interface.

Functional Block Diagram & Bus Interconnects



Data Bus: Transfers raw data/opcodes between CPU, RAM, and I/O devices.
Address Bus: Specifies precise RAM memory locations or I/O port addresses.
Control Bus: Carries READ/WRITE signals, clock synchronizations, and interrupt lines.

Interactive Instruction Cycle & Bus Traffic Simulator

Select Instruction Flow Mode: 1. Fetch Instruction (RAM -> CPU)
Simulate Bus Transaction

Click "Simulate Bus Transaction" to observe real-time address, control, and data signal transfers across hardware buses.

Question 3 (Applying) Campus Scanning & Card Reading

Automated Campus Infrastructure: Scanners & Smart Cards

Question Statement: Demonstrate how optical data scanning devices (e.g., barcode/OMR scanners) and electronic card readers (e.g., RFID/smart card readers) can be deployed in a campus environment to automate student attendance and library checkouts.

Lecture Attendance Automation

13.56 MHz High-Frequency RFID & OMR Backup

- • **Primary Channel (RFID):** Wall-mounted contactless readers at classroom doors scan student ID cards embedded with NTAG213 chips upon entry.

- • **Secondary Backup (OMR):** Optical Mark Recognition scanners parse bubble-sheet exam rosters for offline compliance logging.
- • **Backend Sync:** UID data is pushed over campus Wi-Fi to the Central Academic Server to log timestamped attendance.

Library Self-Checkout & Access

Dual 1D/2D Barcode & RFID Drop Box

- • **Book Cataloging (Barcode):** Linear Code 128 / QR codes scanned via 650nm Laser / CCD Image Scanners at checkout kiosks.
- • **Anti-Theft Gates (UHF RFID):** 860-960 MHz RFID security gates detect un-returned tagged books passing through library exits.
- • **Automated Drop Box:** Instantly clears student accounts upon inserting books into the intelligent drop chute.

Live Campus Smart Terminal Simulator

Simulate Student Action

Tap RFID Student Card `UID: 0x8F92A1`

Scan Library Book Barcode `ISBN: 978-0134685991`

Feed OMR Attendance Sheet `Batch: #409`

```
Terminal Access Log Output
[SYS] Terminal Online. Ready for card tap or barcode scan...
```

Question 4 (Analyzing) Language Taxonomy

Low-Level vs. High-Level Languages Taxonomy

Question Statement: Distinguish between low-level languages (Machine and Assembly) and high-level languages in terms of execution speed, hardware dependence, translation requirements, and programming complexity.

Language Classification Matrix

Evaluation Criteria	Machine Language (1GL)	Assembly Language (2GL)	High-Level Languages (3GL/4GL)
Execution Speed	Instantaneous (Native Binary)	Near-Instant (1:1 Translation)	Slower due to abstraction overhead
Hardware Dependence	100% CPU-Specific (ISA dependent)	100% CPU-Specific	Portable (Platform Independent)

Evaluation Criteria	Machine Language (1GL)	Assembly Language (2GL)	High-Level Languages (3GL/4GL)
Translation Tools	None Required	Assembler	Compiler / Interpreter / JIT
Programming Complexity	Extremely High (0s & 1s)	High (Mnemonic Opcodes)	Low (English-like syntax)
Memory Control	Direct Binary Pointers	Direct Register Allocation	Managed / Garbage Collected

Interactive Code Abstraction Level Comparison

High-Level (Python)

```
c = a + b
print(f"Result: {c}")
```

Assembly (x86_64)

```
MOV EAX, [a]
ADD EAX, [b]
MOV [c], EAX
```

Machine Code (Opcode)

```
8B 05 04 00 00 00
03 05 08 00 00 00
89 05 0C 00 00 00
```

Question 5 (Analyzing) Point & Draw vs. Graphics Digitizers

Point-and-Draw Devices vs. Graphics Digitizers

Question Statement: Compare point-and-draw devices (such as a mouse and trackball) with graphics digitizers/drawing tablets, highlighting how their coordinate tracking mechanisms and pressure sensitivity differ for CAD and digital art applications.

Point-and-Draw (Mouse / Trackball)

- • **Coordinate Tracking:** Relative Motion Vector (ΔX, ΔY). The cursor moves relative to its prior position, requiring re-centering.
- • **Pressure Levels:** Binary (On/Off state click buttons only; 0 or 100% activation).
- • **Hardware Tech:** Optical LED / Laser sensors tracking micro-surface textures.

- • **Primary Use:** Desktop navigation, web browsing, simple UI interactions.

Graphics Digitizer (Drawing Tablet)

- • **Coordinate Tracking:** Absolute Mapping (X, Y). Every point on the active tablet surface maps 1:1 to a screen pixel.
- • **Pressure Levels:** 8,192 Discrete Levels of Stylus Force via Piezoelectric / EMR tip.
- • **Hardware Tech:** Electro-Magnetic Resonance (EMR) grid emitting sensor loops under the tablet.
- • **Primary Use:** Precision CAD drafting, 3D sculpt modelling, digital painting, vector illustration.

Interactive Digitizer Pressure Simulator

Stylus Pressure Force: 4096 / 8192

Stylus Tilt Angle: 45°

Clear Canvas

Question 6 (Evaluating) Plotter vs. UHD + Printer Trade-offs

Technical Evaluation: Plotters vs. UHD Monitor & Large-Format Printer

Question Statement: Assess whether a drum/flatbed plotter or an ultra-high-definition monitor and large-format printer combination is superior for producing civil engineering blueprints, justifying your selection based on line resolution, scalability, and long-term production costs.

Technical Justification & Verdict

Legacy Vector Plotters (Drum / Flatbed)

Draws continuous vector lines using mechanical ink pens. While offering zero raster aliasing, speed drops drastically on complex multi-layer prints. Mechanical parts (pen arms, motors) suffer high wear and tear.

Modern Solution: 4K/8K UHD Monitor + Large-Format Inkjet Printer

SUPERIOR CHOICE: High DPI thermal/piezoelectric inkjet heads print complete 36"–44" architectural sheets in seconds regardless of line complexity. UHD monitors allow real-time CAD editing with zero ink waste.

Performance & Cost Evaluation

Print Speed on Complex BIM Blueprints UHD + Printer (5x Faster)
Multi-Layer Color Shading & Render Support UHD + Printer (100% Support vs Pen Plotter Line-Only)

Question 7 (Evaluating) Biometric Security Analysis

Vision vs. Voice Biometric Authentication in Adverse Environments

Question Statement: Critique the operational reliability and security vulnerabilities of vision-input devices (facial recognition) compared to voice recognition devices when deployed as primary biometric authentication tools in high-ambient-noise, publicly exposed environments.

Adverse Environmental Impact Simulation

Environmental Controls

Ambient Noise Level: 85 dB
Lighting / Glare: Low Light / Glare

Authentication Failure Rate Output

Voice Recognition Error Rate 78%
Vision (Facial + IR) Error Rate 22%

Question 8 (Creating) Assistive Workstation Synthesis

Accessible Hands-Free Workstation for Visually Impaired Users

Question Statement: Design an accessible, hands-free computer workstation tailored for a visually impaired user. Propose a coherent integration of non-visual input devices (such as voice recognition systems and tactile keyboards) with specialized output devices (such as voice response synthesizers and screen readers).

Integrated Non-Visual Inputs

- • **Array Beamforming Microphones:** Hands-free voice recognition with noise cancellation for dictation & OS commands.

- • **Tactile Braille Keyboard:** High-contrast Perkins-style tactile chording keys with physical guides.
- • **Foot Pedal Switches:** Programmable foot switches mapped to modifier keys (Shift, Ctrl, Voice Trigger).

Non-Visual Tactile & Audio Outputs

- • **Refreshable Braille Display:** Piezoelectric 80-cell tactile pin array for real-time document reading.
- • **Text-To-Speech Synthesizer:** NVDA / JAWS screen reader with spatial 3D audio feedback.
- • **Haptic Wristbands:** Micro-vibrations signaling system alerts, errors, and navigation cues.

Interactive 6-Dot Refreshable Braille Cell Output

Type Text to Translate to Braille: HELLO Speak Text (TTS)

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